Square-first pair normalizer, signed score transfer, and filtration boundary

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1 Purpose and scope

R-108 reduced the active A13 theorem to a cutoff-, chart-, control-, and subdivision-uniform signed complete-cluster lower bound, or to a legal square-before-average conditional normalizer which implies it. On the smallest contraction-closed Fourier pair, R-108 established only a small- amplitude expansion and one finite viability check. The present package turns that viability check into an all-amplitude theorem and then audits exactly which parts survive same-root feedback.

The result has six parts.

1. A conditional one-sided Bennett lemma converts a centered lower floor and a variance-to-floor ratio into a global quadratic log-Laplace cost. No smallness assumption remains.
2. The complete one-pair cluster $\{0, +2, -2\}$ satisfies that lemma for every predictable scale, every nonnegative weight, and every $q \geq 0$. Its exact cost is bounded by one quarter of the conditional expectation of the squared realized whole-cluster covariance. Legal fresh pairs therefore iterate to a supermartingale.
3. The R-108 target schema is repaired at the filtration level. A covariance revealed by the current root cannot appear raw on the right of an $\mathcal{F}_{j-1}$ conditional normalizer. The legal cost is its conditional expectation, or a proved predictable envelope.
4. The pure quartic pair has a selector-independent pathwise floor whose production diagonal sum is cutoff-uniform. The complete pair's baseline floor nevertheless diverges; its determinant conversion is indispensable, and physical cross-mode resonances prevent pairwise tensorization.
5. A complete signed second jet with one past-measurable covariance operator admits exact Gaussian score transfer and a source/sextic form bound. This proves uniform control of the signed oscillatory companion left open by R-108, but only for that fixed-operator owner.
6. Gaussian score transfer is expectation-only. An exact oscillatory family has a derivative-representative log-Laplace cost growing linearly while the raw Wick coordinate tends to zero exponentially in the relevant Orlicz sense. Thus the derivative representative may not be substituted inside a conditional exponential normalizer.

These are genuine analytic advances and one target correction. They do not identify the full adapted production forest with one predictable fixed- operator second jet, control cross-mode same-root feedback, prove the uniform complete-cluster bound, establish $\mathrm{OVERLAP}_{\mathrm{src}}$ or Nelson, remove any cutoff or floor, construct an interacting measure, close Sector A, or change the tier from T4.

2 A conditional one-sided normalizer lemma

Theorem 2.1 (conditional floor–variance normalizer)

Let $\mathcal{F}$ be a sigma algebra. Suppose Z is real, $\beta \ge 0$ is $\mathcal{F}$-measurable, and conditionally on $\mathcal{F}$,

$$\mathbb{E}[Z \mid \mathcal{F}] = 0, \qquad Z \le \beta, \qquad \mathbb{E}[Z^2 \mid \mathcal{F}] \le 8\beta^2. \tag{2.1}$$

Then for every $q \ge 0$,

$$\boxed{\log \mathbb{E}[e^{qZ} \mid \mathcal{F}] \le 5q^2\beta^2.} \tag{2.2}$$

Proof. The event $\{\beta = 0\}$ is immediate. Else set $X = Z/\beta \le 1$ and $a = q\beta$. For every $x \le 1$,

$$e^{ax} \le 1 + ax + (e^a - 1 - a)x^2. \tag{2.3}$$

For $0 \le x \le 1$, (2.3) follows term by term from the exponential series. For $x < 0$, use $e^{ax} \le 1+ax+a^2x^2/2$ and $a^2/2 \le e^a - 1 - a$. Taking conditional expectations gives

$$\log \mathbb{E}e^{aX} \le 8(e^a - 1 - a). \tag{2.4}$$

If $0 \le a \le 1/5$, then $n! \ge 2\,3^{n-2}$ for $n \ge 2$, so

$$e^a - 1 - a \le \frac{a^2}{2(1-a/3)}, \qquad 8(e^a - 1 - a) \le \frac{30}{7}a^2 < 5a^2. \tag{2.5}$$

If $a \ge 1/5$, the pointwise bound $X \le 1$ gives

$$\log \mathbb{E}e^{aX} \le a \le 5a^2. \tag{2.6}$$

The two intervals cover $[0, \infty)$. $\square$

The constants are deliberately compatible with the realized covariance of the production one-pair fixture. The infinitesimal coefficient in that fixture is 4, so a universal coefficient below 4 is impossible. The coefficient 5 has one unit, or twenty percent, of leading room.

3 The all-amplitude complete one-pair theorem

Let $\mathcal{F}_{j-1}$ be the strict past. Assume $\lambda, \sigma \ge 0$ are $\mathcal{F}_{j-1}$-measurable and, conditionally on the past, ξ, η are independent standard Gaussians. Put

$$z = \frac{\sigma}{2}(\xi - i\eta), \qquad R = |z|^2 = \frac{\sigma^2}{2}T, \qquad T \sim \mathrm{Exp}(1), \tag{3.1}$$

and retain the complete output cluster $\{0, +2, -2\}$. Its diagonal packet is

$$P_{\lambda,\sigma} = \lambda(R^2 - \sigma^2 R) = \frac{\lambda\sigma^4}{4}(T^2 - 2T). \tag{3.2}$$

Theorem 3.1 (all-amplitude square-before-average normalizer)

Let S_z^λ be the R-107/R-108 realized independent-copy covariance of the whole cluster after the current is scaled by $\sqrt{\lambda}$. Then for every $q \ge 0$,

$$\boxed{\log \mathbb{E}[e^{-qP_{\lambda,\sigma}} \mid \mathcal{F}_{j-1}] \le \frac{q^2}{4}\mathbb{E}[\|S_z^\lambda\|_{\mathfrak{S}_2}^2 \mid \mathcal{F}_{j-1}].} \tag{3.3}$$

Proof. Set

$$\beta = \frac{\lambda\sigma^4}{4}, \qquad U = 2T - T^2 = 1 - (T-1)^2. \tag{3.4}$$

Then $-P = \beta U$, $U \le 1$, and the exact exponential moments $\mathbb{E}T^n = n!$ give

$$\mathbb{E}U = 0, \qquad \mathbb{E}U^2 = 8. \tag{3.5}$$

Theorem 2.1 yields

$$\log \mathbb{E}e^{-qP} \le 5q^2\beta^2. \tag{3.6}$$

The two nonzero eigenvalues of S_z^λ are

$$\frac{3}{2}\lambda\sigma^2 R, \qquad \frac{1}{2}\lambda\sigma^2 R. \tag{3.7}$$

Since $\mathbb{E}R^2 = \sigma^4/2$,

$$\mathbb{E}\|S_z^\lambda\|_{\mathfrak{S}_2}^2 = \frac{5}{4}\lambda^2\sigma^8, \tag{3.8}$$

and the right side of (3.3) is exactly

$$\frac{5q^2\lambda^2\sigma^8}{16} = 5q^2\beta^2.$$

□

This theorem upgrades the R-108 small-σ expansion and $(q,\sigma) = (1, 1/2)$ check to every amplitude. Indeed

$$\log \mathbb{E}e^{-qP} = 4a^2 - 40a^3 + O(a^4), \qquad a = \frac{q\lambda\sigma^4}{4}, \tag{3.9}$$

while the square-first cost is $5a^2$.

Corollary 3.2 (legal sequential fresh-pair product)

Suppose P_j has the form (3.2) conditionally on $\mathcal{F}_{j-1}$ and define the predictable cost

$$C_j = \frac{q^2}{4}\mathbb{E}[\|S_j^{\mathrm{real}}\|_{\mathfrak{S}_2}^2 \mid \mathcal{F}_{j-1}]. \tag{3.10}$$

Then

$$M_n = \exp(-q\sum_{j\le n} P_j - \sum_{j\le n} C_j) \tag{3.11}$$

is a nonnegative supermartingale. This is the legal triangular iteration which the global frozen future-row determinant of R-107 cannot replace.

The hypotheses are exact: the pair is fresh; its scale and nonnegative weight are strict-past measurable; every contraction-connected output remains in one cluster; and no same-root selector, cross-mode resonance, or rational coefficient variation is hidden in P_j. No injectivity of the rectangular root-to-output row is required. The identity operator in the frozen determinant makes singular covariance harmless.

4 Filtration repair of the R-108 target schema

Audit 4.1 (square first, then condition)

R-108 (8.3) is a target schema, not an asserted determinant. Its right-hand cost must be measurable at the conditioning step. If $S_{j,\mathcal{C}}$ is the covariance of a strict-past affine row, then $\|S_{j,\mathcal{C}}\|_{\mathfrak{S}_2}^2$ is already $\mathcal{F}_{j-1}$-measurable and the displayed R-108 form is legal.

If instead $S^{\mathrm{real}}_{j,\mathcal{C}}$ depends on the root being integrated, the legal square-before-average target is

$$\begin{aligned}\log \mathbb{E}\left[e^{-q(\mathfrak{P}_{j,\mathcal{C}}+\text{payments})} \mid \mathcal{F}_{j-1}\right] \\ \le \frac{q^2}{4}\mathbb{E}[\|S^{\mathrm{real}}_{j,\mathcal{C}}\|^2_{\mathfrak{S}_2} \mid \mathcal{F}_{j-1}] + r^{\mathrm{pred}}_{j,\mathcal{C}},\end{aligned} \tag{4.1}$$

where $r^{\mathrm{pred}} \ge 0$ is $\mathcal{F}_{j-1}$-measurable, or the conditional cost is replaced by an explicitly proved predictable envelope. Equation (4.1) keeps the square inside and the averaging after, as R-108 Proposition 6.1 requires. It remains a target for a general production cluster.

There is a subtle but essential two-stage boundary. After revealing the root, an auxiliary independent copy gives the frozen R-107 determinant

$$\log \mathbb{E}_w e^{-q(\|A_z w\|^2 - \operatorname{Tr} S_z)/2} \le \frac{q^2}{4}\|S_z\|^2_{\mathfrak{S}_2}. \tag{4.2}$$

Returning to $\mathcal{F}_{j-1}$ generally produces

$$\log \mathbb{E}\left[\exp\left\{\frac{q^2}{4}\|S_z\|^2_{\mathfrak{S}_2} + r_z\right\} \mid \mathcal{F}_{j-1}\right], \tag{4.3}$$

not the cheaper conditional mean in (4.1). Moreover, (4.2) normalizes the auxiliary decoupled packet, not the original diagonal current. As $R \downarrow 0$, the original negative exponent is linear in R, whereas the determinant cost in (4.2) is quadratic in R; no pointwise comparison exists. Theorem 3.1 is the separate diagonal-cluster lemma which justifies (4.1) in the one-pair case.

This audit does not withdraw any exact R-108 endpoint, covariance, quotient, or order-of-operations result. It makes the target's admissible measurability interpretation explicit.

5 Selector-independent quartic floor and its exact limitation

Proposition 5.1 (arbitrary same-root selector floor)

For every $R \ge 0$, $\sigma \ge 0$, and nonnegative weight λ, without any Gaussianity, independence, differentiability, or filtration assumption,

$$\boxed{\lambda(R^2 - \sigma^2 R) = \lambda\left(R - \frac{\sigma^2}{2}\right)^2 - \frac{\lambda\sigma^4}{4} \ge -\frac{\lambda\sigma^4}{4}.} \tag{5.1}$$

Thus any sequential family satisfies

$$\sum_{j\le J}\lambda_j P_j \ge -\frac{1}{4}\sum_{j\le J}\lambda_j\sigma_j^4. \tag{5.2}$$

If the right side is deterministically bounded, (5.2) gives a uniform direct and exponential bound. If it is random, only $\log \mathbb{E}\exp(q\sum \lambda_j\sigma_j^4/4)$ is justified; its expectation is not enough.

For the artificial R-105 diagonal pair, put

$$R = \frac{\sigma_k^2}{4}S_h, \qquad \mu_k = \frac{\beta_0 k^2 \sigma_k^4}{16}. \tag{5.3}$$

Then

$$\mu_k(S_h^2 - 4S_h) = \beta_0 k^2 (R^2 - \sigma_k^2 R) \tag{5.4}$$

has floor $-4\mu_k$. Since $\sigma_k^2 \lesssim \langle k\rangle^{-4}$ in three dimensions,

$$\sum_k 4\mu_k \lesssim \sum_k \langle k\rangle^{-6} < \infty. \tag{5.5}$$

The shell decay is 2^{-3j}. Hence the pure centered quartic diagonal submodel is cutoff-uniform even for an arbitrary same-root selector.

Proposition 5.2 (the full direct pair floor still diverges)

The complete artificial pair of R-105 is

$$V(S) = \alpha(S-2) + \mu(S^2 - 4S), \qquad S \geq 0. \tag{5.6}$$

For $\mu > 0$ its exact minimum is

$$\inf_{S\geq 0} V(S) = \begin{cases} -2\alpha, & \alpha \geq 4\mu, \\ -4\mu - \alpha^2/(4\mu), & 0 \leq \alpha < 4\mu. \end{cases} \tag{5.7}$$

For $\mu = 0$ the first line remains. In R-105,

$$\frac{\alpha_k}{\mu_k} = \frac{4A^2}{\sigma_k^2} \longrightarrow \infty \quad (A \neq 0), \tag{5.8}$$

so the high modes are forced onto the $-2\alpha_k$ branch, and

$$\sum_k \alpha_k \asymp A^2 \sum_k \langle k \rangle^{-2} = \infty. \tag{5.9}$$

The determinant changes this nonsummable linear loss to $t - \log(1+t) = O(t^2)$, whose shell cost is 2^{-j} and is summable.

The physical action cannot be closed by multiplying these pair estimates. R-105 (8.10)–(8.11) gives a sign-indefinite k:$2k$ resonance, while R-107 requires contraction-connected outputs, covariance blocks, mixed baseline, trace, and forest owners to remain in one atom. Equations (5.5) and (5.7) are therefore a positive diagonal submodel theorem and an exact limitation, not a production cluster decomposition.

6 Signed second-jet score transfer

Let $\mathcal{F}_{\mathrm{past}}$ be fixed. Conditionally on it, let $\xi \sim N(0, \mathrm{I}_d)$ be fresh, let $W = W^* \succeq 0$ be finite-rank and $\mathcal{F}_{\mathrm{past}}$-measurable, and let $h = h(\mathcal{F}_{\mathrm{past}}, \xi)$ be a twice differentiable cylindrical map into a real Hilbert space. Assume the displayed terms are integrable and define

$$\begin{aligned}\mathfrak{F}_W(h) = \mathbb{E}_\xi \big[&\| D_\xi h \, W^{1/2} \|_{\mathfrak{S}_2}^2 \\ &+ \langle h, \mathrm{Tr}_{\mathbb{R}^d}(W D_\xi^2 h) \rangle \mid \mathcal{F}_{\mathrm{past}} \big].\end{aligned} \tag{6.1}$$

Theorem 6.1 (exact score transfer and form bound)

One has the exact identity

$$\boxed{\mathfrak{F}_W(h) = \frac{1}{2} \mathbb{E}_\xi \left[(\xi^T W \xi - \mathrm{Tr}\, W) \|h\|^2 \mid \mathcal{F}_{\mathrm{past}} \right].} \tag{6.2}$$

Writing

$$X = \mathbb{E}_\xi [\|h\|^2 \mid \mathcal{F}_{\mathrm{past}}], \qquad Y = \mathbb{E}_\xi [\|h\|^6 \mid \mathcal{F}_{\mathrm{past}}], \tag{6.3}$$

gives

$$\boxed{|\mathfrak{F}_W(h)| \leq \frac{\|W\|_{\mathfrak{S}_2}}{\sqrt{2}} X^{1/4} Y^{1/4}.} \tag{6.4}$$

Consequently, for every $q, \eta, \zeta > 0$,

$$\boxed{-\frac{q}{2} \mathfrak{F}_W(h) \geq -\eta X - \zeta Y - \frac{q^2 \|W\|_{\mathfrak{S}_2}^2}{64\sqrt{\eta\zeta}}.} \tag{6.5}$$

Proof. The Hilbert chain rule gives

$$\frac{1}{2}D^2\|h\|^2 = (Dh)^*Dh + \langle h, D^2h\rangle. \tag{6.6}$$

Contract (6.6) with W and apply Gaussian integration by parts twice. This proves (6.2). Also

$$\mathbb{E}_\xi(\xi^T W\xi - \operatorname{Tr} W)^2 = 2\|W\|^2_{\mathfrak{S}_2}. \tag{6.7}$$

Cauchy–Schwarz and interpolation $\mathbb{E}\|h\|^4 \le X^{1/2}Y^{1/2}$ give (6.4). Finally

$$Ax^{1/4}y^{1/4} \le \eta x + \zeta y + \frac{A^2}{8\sqrt{\eta\zeta}} \tag{6.8}$$

is sharp at

$$x = \frac{A^2}{16\eta^{3/2}\zeta^{1/2}}, \qquad y = \frac{A^2}{16\eta^{1/2}\zeta^{3/2}}. \tag{6.9}$$

Use $A = q\|W\|_{\mathfrak{S}_2}/(2\sqrt{2})$. □

At $q = 10/9$, the coefficient $q^2/64$ in (6.5) is exactly $25/1296$. For an R-063 derivative-root covariance shell,

$$W_{i,j}(k) = \mathbf{1}_{\{|k|\sim 2^j\}}k_i^2 r_\Lambda(k)^2\widehat{C}(k), \tag{6.10}$$

the A1 bound $\|\widehat{C}(k)\|_{\mathrm{op}} \lesssim \langle k\rangle^{-4}$ gives

$$\|W_{i,j}\|^2_{\mathfrak{S}_2} \lesssim \sum_{|k|\sim 2^j} \langle k\rangle^{-4} \lesssim 2^{-j}. \tag{6.11}$$

Thus a genuinely fixed, past-measurable W has cutoff-uniform HS cost after complete aggregation.

The missing production identification is load-bearing. R-063 proves the deterministic-shift reconstruction of the $P_3/P_1/P_4/P_2/\Sigma Q/P_0$ forest. R-107 and R-108 do not prove for arbitrary same-root future feedback that the whole contraction-closed trace defect equals one $\mathfrak{F}_{W_\mathcal{C}}(h_\mathcal{C})$ with $W_\mathcal{C}$ past measurable, nor that its X, Y use the canonical source energy and terminal sextic without visit multiplicity. If $W = W(\xi)$, integration by parts also differentiates W and creates the unmapped DW, D^2W forest terms.

Corollary 6.2 (the R-108 oscillatory signed owner closes)

For $d = 1$, $W = 1$, and $h_M(G) = a\sin(MG)$,

$$\mathfrak{F}_1(h_M) = \mathbb{E}\{(h_M')^2 + h_M h_M''\} = a^2M^2e^{-2M^2}. \tag{6.12}$$

Thus Theorem 6.1 uniformly controls the complete signed second-jet owner, not merely its particular asymptotic cancellation. It does not control the positive tangent square, its rank-one HS square, the mixed current, or the full complete packet, so the R-108 absolute-ledger no-go remains intact.

7 Score transfer may not be exponentiated

Proposition 7.1 (expectation cancellation is not exponential cancellation)

Let $G \sim N(0,1)$ and scale

$$h_M(G) = M^{-1/2}\sin(MG). \tag{7.1}$$

The derivative representative is

$$K_M = (h_M')^2 + h_M h_M'' = M\cos(2MG), \qquad \mathbb{E}K_M = Me^{-2M^2}. \tag{7.2}$$

For every fixed $\theta \neq 0$, the Fourier–Bessel expansion gives

$$\mathbb{E}e^{\theta M \cos(2MG)} = I_0(\theta M) + 2\sum_{n\geq 1} I_n(\theta M)e^{-2n^2M^2}. \tag{7.3}$$

Therefore

$$\boxed{\log \mathbb{E}e^{\theta(K_M - \mathbb{E}K_M)} = |\theta|M - \frac{1}{2}\log(2\pi|\theta|M) + o(1).} \tag{7.4}$$

The centered derivative-representative log-Laplace cost diverges linearly.

The raw Wick coordinate on the other side of the Stein identity is

$$W_M = h_M(G)^2(G^2 - 1), \qquad \mathbb{E}W_M = 2\mathbb{E}K_M. \tag{7.5}$$

But

$$|W_M| \leq \frac{G^2+1}{M}, \tag{7.6}$$

and for $M > 2|\theta|$,

$$\boxed{\log \mathbb{E}e^{|\theta||W_M|} \leq \frac{|\theta|}{M} - \frac{1}{2}\log\left(1 - \frac{2|\theta|}{M}\right) \longrightarrow 0.} \tag{7.7}$$

Equations (7.4)–(7.7) prove that the expectation identity (6.2) cannot be used as a pathwise substitution inside a conditional exponential estimate. A determinant or log-Laplace proof must remain in the raw Wick and complete- endpoint coordinate until after the exponential step, unless a separate pathwise full-cluster identity is proved. This is not a counterexample to the direct expectation bound (6.5), Theorem 3.1, the raw complete packet, or the Nelson objective.

8 Why the remaining mean/covariance debt is not closed

R-108 (4.6) is already sharp enough to exclude a PSD-only completion. Put $B_0 = B_1 = B \succeq 0$ and $V = \Gamma$. The $q = 10/9$ optimized mean debt becomes

$$-\frac{5}{9}\mu^T B^2(B + \frac{9}{10}\mathrm{I})^{-1}\mu. \tag{8.1}$$

On a B-eigenvector of eigenvalue λ, its coefficient is

$$-\frac{5}{9}\frac{\lambda^2}{\lambda + 9/10} \sim -\frac{5}{9}\lambda. \tag{8.2}$$

Thus common deterministic heat and PSD alone provide no fixed source coefficient. The covariance debt also has no sign because positivity gives neither $B_1 \succeq B_0$ nor $V \succeq \Gamma$.

Gaussian Poincare has the wrong orientation:

$$\mathrm{Cov}(Y) \preceq \mathbb{E}[DY(DY)^*], \tag{8.3}$$

whereas R-108 needs a lower estimate on $\mathrm{Cov}(Y) - T$. Theorem 6.1 repairs one signed fixed-W second jet only after the two Hessian companions are summed. It does not couple (8.1), the random covariance defect, mixed current, rational recovery, heat, visits, and the full R-063 forest.

The exact remaining production object is therefore narrower than R-108 but still nontrivial: prove a diagonal-to-decoupled or direct signed estimate for one contraction-closed same-root cluster in raw endpoint coordinates, with the conditional covariance cost predictable, every random-W derivative forest retained, and canonical source/sextic domination without multiplicity.

9 Proof-route evidence map

Route	Verdict	Reusable reason
One-pair square-first normalizer	proved	Conditional Bennett plus the exact whole-cluster covariance gives the cost for all amplitudes.
Sequential fresh-pair product	proved	The conditional costs are predictable, so the normalized exponentials form a supermartingale.
Raw realized cost at a pre-root step	repaired	Use the conditional expectation of the squared realized cost or a predictable envelope.
Auxiliary-copy determinant for the diagonal packet	failed	It normalizes a decoupled packet and produces an exponential of the random cost one level earlier.
Arbitrary-selector pure quartic floor	proved in the diagonal submodel	The floor mass is $O(\langle k\rangle^{-6})$ per mode.
Full pair direct floor	failed	The high-frequency baseline is forced onto the nonsummable $-2\alpha_k$ branch.
Fixed-W signed second jet	proved	Two Gaussian integrations by parts and one exact Young optimization give (6.5).
Stein derivative inside a normalizer	failed	Its log-Laplace cost is linear in M, while the raw Wick coordinate tends to zero.
PSD, common heat, or Poincare alone	failed as a closure	The mean debt is negative and linearly unbounded; Poincare has the wrong sign.
Full production complete cluster	open	Cross-mode, same-root random-W, mixed, covariance, rational, visit, and forest owners remain coupled.

These are researcher-reusable route decisions, not private search traces. Their exact questions, finite checks, boundaries, and revisit conditions are also recorded in the append-only exploration ledger.

10 Devil's-advocate audit

1. **Objection: R-108 already proved the one-pair determinant.**

   UPHELD AS A SCOPE CORRECTION. R-108 proved the leading expansion, one finite value, and viability of the square-first coefficient. Theorem 3.1 is the first global-in-amplitude inequality for the actual diagonal packet.

2. **Objection: the raw realized covariance can remain on the right because it is eventually averaged.**

   UPHELD AGAINST THE CLAIM. A pre-root conditional logarithm is $\mathcal{F}_{j-1}$-measurable. A root-dependent raw cost is not a legal iterable normalizer. Equation (4.1) is the corrected target.

3. **Objection: reveal the root, use an independent copy, and then average the determinant cost.**

   UPHELD AGAINST THE SHORTCUT. The outer operation is the logarithm of an exponential moment of the random cost, as in (4.3), and the auxiliary packet is not the original diagonal packet.

4. **Objection: the selector-independent quartic floor closes the physical production action.**

   UPHELD AGAINST THE CLAIM. The full pair baseline has the divergent $-2\alpha_k$ branch, and physical $k{:}2k$ resonances prohibit multiplication of isolated pair floors.

5. **Objection: Theorem 6.1 overturns the R-108 future-feedback Carleson no-go.**

   DISMISSED. The theorem controls only the signed second-jet sum with one predictable W. R-108 refutes the separated positive tangent square and its HS square, which remain unbounded in that architecture.

6. **Objection: once score transfer is exact, the Wick forest can be replaced before the exponential estimate.**

   UPHELD AGAINST THE CLAIM. Proposition 7.1 gives a linear versus vanishing log-Laplace separation. Score transfer is expectation-only.

7. **Objection: PSD and Gaussian Poincare finish the remaining covariance debt.**

   UPHELD AGAINST THE CLAIM. Equations (8.1)–(8.3) give the exact negative mean coefficient and the wrong Poincare orientation.

11 Authorities and scope firewall

The load-bearing authorities are R-063 (balanced forest reconstruction), R-077 (fresh-root complete-packet cancellation), R-082 (coherent output and covariance boundary), R-104 (temporally faithful source charts), R-105 (artificial one-pair determinant and cross-mode boundary), R-107 (fresh-root whole-output determinant and same-root residual), and R-108 (complete-cluster normal form and covariance-order audit).

Proved here are Theorem 2.1, the all-amplitude complete one-pair normalizer, its fresh-pair supermartingale, the filtration audit, the arbitrary-selector pure quartic floor and exact full-pair floor split, the fixed-predictable-W score-transfer theorem, its oscillatory corollary, and the exact Stein-exponentiation no-go.

Not proved here are a tensorization across physical Fourier pairs, the diagonal-to-decoupled theorem for a general contraction-connected cluster, the adapted random-W R-063 forest identification, a uniform complete source-action lower bound, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, removals, an interacting measure, Sector-A closure, or a T5–T7 promotion.

12 Executable verification

The primary certificate derives the exponential and Gaussian moments, Bennett split, covariance eigenvalues, floor minima, score-transfer constants, shell power counts, oscillatory asymptotics, CM debt, and filtration dependencies. The independent certificate imports neither the primary nor symbolic/scientific array libraries; it repeats the exact arithmetic with standard-library fractions and finite exact-log-mgf checks. The integrated verifier pins both outputs, all authorities, the note, PDF, manifest, audit and negative records, exploration decisions, public ledgers, and scope firewall.

Run

```
E:\Dev\TECT.venv\Scripts\python.exe ‘
codes/foundations/a13_classii_square_first_pair_score_transfer_filtration_boundary_verify.py
```

Result footer

```
Result ID: A13-CLASSII-SQUARE-FIRST-PAIR-SCORE-TRANSFER-FILTRATION-BOUNDARY
Claim:     A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Ledger ID: R-109
Status:    T4 ANALYTIC/EXACT/EXECUTED ADVANCE AND METHOD BOUNDARY;
           NELSON AND SECTOR A OPEN
Scope:     Finite cutoff, fixed positive floor; exact all-amplitude fresh-pair
           normalizer, conditional filtration audit, diagonal selector floor,
           and fixed-predictable-W signed second-jet score transfer.
Precise statement: the complete one-pair cluster obeys the square-first
           conditional cost for all q and amplitudes; the R-108 root-realized
           cost must be conditionally averaged before use at the prior step;
           one complete fixed-W second jet has a direct form bound; Stein
           differentiation cannot be substituted inside a log-Laplace bound.
Dependencies: R-063, R-077, R-082, R-104, R-105, R-107, R-108.
Evidence grade: ANALYTIC/EXACT/EXECUTED; TWO NON-IMPORTING ROUTES.
Reproduction command:
           E:\Dev\TECT.venv\Scripts\python.exe ‘
           codes/foundations/a13_classii_square_first_pair_score_transfer_filtration_boundary_verify.py
Expected output:
           primary 39/39; independent 93/93; integrated count manifest-pinned;
           aggregate count manifest-pinned; PASS.
Negative/audit records:
           AUDIT-2026-07-28-A13-R108-REALIZED-COVARIANCE-FILTRATION;
           NG-2026-07-28-A13-STEIN-SECOND-JET-EXPONENTIATION.
Falsification gate: failure of any Bennett, covariance-square, floor,
           score-transfer, Young, shell, oscillatory, filtration, authority,
           hash, PDF, ledger, count, or scope assertion.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement:
           the full adapted production cluster, OVERLAP_src, Nelson, removals,
           measure, and Sector A remain open.
Next required action: prove the raw complete same-root diagonal-to-decoupled
           or direct signed cluster estimate with predictable square cost,
           random-W derivative forest, cross-mode baseline/current, and one-
           use source/sextic domination retained together.
```